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1 Authority, scientific objects and compatible arithmetic

Program adherence and prior-work recovery. The r0.3 fixed-operator repair and three-document structure are the accepted revision basis. This is the complete sole current Stage B review-candidate normative core, superseding r0.3 as the candidate submitted for review. Prior artifacts are exact historical input and comparison evidence; no unstated historical sentence supplies a current obligation. Exact Stage A sources, preserved r0.2/r0.3 deliveries and the r0.4 directive bind provenance. Only the admitted analysis/gridding-coefficient ownership and route-use boundary is restored. Accepted response, uncertainty, support, lifecycle and terminal rules remain unchanged; no broad derivation or Stage A reopening occurs. This candidate is neither owner approved nor frozen, registered or activated. Product names below are core declarations, not claims of an actual FRUIT Registry record. **A** (program order, decision 2026-09-06) authorizes conditional authorship; **B** (conditional composition, decision 2026-09-06) authorizes conditional remove/process/rejoin consequences. **C**, FRUIT-FEEDBACK-METHOD v0.1, is a required-record family, not an approved numerical instance. A/B do not imply C. Every numerical route remains `unavailable_pending_separate_owner_approval`. Stage A is frozen; Stage B is an owner-review draft. Exact decisions and preserved bytes are in the authority packet.

Each scientific object binds quantity, unit/numerical scale, frame/domain, grid/sampling, shape/axes/indexing, timing where relevant, grouping, support, validity and missing/non-finite causes, producer/parent and generations. Shape, time, filename or a shared label does not establish identity. Bind separately

$$m_k^{\text{candidate}} \in M_k^{\text{candidate}}, \quad m_k^{\text{accepted}} \in M_k^{\text{accepted}}, \quad m_k^{\text{applied}} \in M_k = M_k^{\text{applied}}, \quad y_k^{\text{in}} \in Y_k, \quad z_k \in Z_k.$$

Candidate is the proposed object; accepted is selected for feedback; applied is the exact removal/rejoin operand. Every transformation among their spaces and objects is explicit, including normalization, clipping, sign conversion, support restriction, projection and unit conversion. Equality needs an exact method/state fact: no equal domain, support, unit, normalization, sign, representation, response, uncertainty or parentage is assumed. External models retain parent/prior; learned models retain causal learning population, selection, response, uncertainty and generations. Acceptance confers only `feedback_model_state`, not truth, a detection, posterior, unbiased sky/source estimate, catalogue, Pointing or OOF result.

Y_k, Z_k shall be exact additive quantity spaces, or explicit affine/quotient spaces with one declared reference/gauge for each operation and well-defined additive-coordinate or quotient arithmetic. Arbitrary representative arithmetic is forbidden. Removal and rejoin *separately* require compatible quantity, unit/scale, frame/domain, grid/sampling, support, additive origin/reference or gauge, null-space state, calibration and response conventions. Every conversion and alignment is declared. Pairwise compatibility does not imply $Y_k = Z_k$. Missing compatibility makes the affected arithmetic and dependent claim unavailable.

Use ρ_k for residual, F_k for the complete residual procedure and S_k for state. New FRUIT notation does not borrow generic x/r , bare Q_p, N_p, C_p or unqualified NOI member/sign symbols. KID x/r retain their paired-readout meaning. The frozen PTC quotation has its own covered local symbols.

2 Ownership and inherited producer boundaries

FRUIT's boundary starts at the exact iteration input and state, with immutable original measured-parent ancestry. It includes model construction/selection/ acceptance/application, removal, residual processing, exact bypass, rejoin, successor state, completion and terminal selection. It ends at immutable iteration or selected-terminal products. The target, measured signal, feedback model, path contributions and result remain different scientific objects.

ALIGN/AST retain coordinate/event meaning; RTC/CAL retain conditioning and calibration; PTC retains its cleaning operators, analysis/gridding-coefficient families and fitted/resolved/applied state; MAP/JINC retain estimands, normalization, grouping, support and numerical admission; NOI retains ensemble/uncertainty authority. Other producers retain their exact product and transformation authority. FRUIT owns its internal selection, update, transition, convergence/stopping, local completeness, terminal selection and publication. It may bind an upstream operation's placement and generation; it shall not redefine that producer's estimator.

PTC ownership and application crosswalk. The exact admitted PTC_APPLICATION_REFERENCE.tex (SCI-PTC v0.1/r0.5) is preserved verbatim. PTC owns its estimator and exact fixed-state application map. FRUIT owns only the response role assigned to a declared use of that upstream operation inside a FRUIT method. The following specialization substitutes the admitted equations; it changes no estimator, family, rank, support or upstream admission.

PTC-local g, t mean detector group/time sample. Its local centered row x , output row z and modal application coordinate $\hat{m}^{\text{app}}$ are PTC objects, distinct from KID readout, FRUIT result z_k and feedback model m_k^{applied} . The loading matrix, metric and normal matrix define the PTC map; the modal coordinate is evaluated from its operand. Ordinary fixed-subspace projection, frozen numerical-component subtraction and relearning the defining state remain separate procedures. The exact map and domain below determine the FRUIT response classification; an internal application coefficient is not an operator-defining coefficient.

Every required input and time-local normal matrix must be finite, with numerical rank equal to the required rank under frozen tolerance. Deficiency makes that group-time application/output unavailable with cause. A generalized inverse permits no partial-rank subtraction, rank reduction, interpolation or borrowed group. Exact output support controls publication. No mode count, metric, projection orientation/family or learned state is selected here. Total removed signal is input minus output, not just the fitted modal contribution. Fixed-state linear null, re-estimated-centering invariance/unidentifiability and affine reference loss remain distinct.

Each parent route supplies exact quantity, unit, frame, response, beam/template, calibration, support/validity, lifecycle and provenance. If the ordinary PTC-to-MAP route is selected, retain its calibrated- x -derived nonpolarimetric total-intensity-equivalent, top-of-atmosphere point-source-equivalent mJy/beam, fixed nominal beam/template and spectral/calibration lineage. This implies neither Stokes I, true sky, a generic FRUIT target nor a universal unit. A STOKES label supplies no authority. A selected MAP/JINC parent route retains exact PTC transformed-output identity, the analysis/gridding-coefficient and QC identities required by that route, and compatible coordinate association. Row/time labels do not establish the same processed occurrence. Internal modal application coordinates remain separate and do not substitute for those handoff facts. Permission for MAP to use a PTC analysis/gridding-coefficient family does not imply permission for JINC to use that family.

JINC remains a signed normalized observation-map estimator; its base authority supplies no MAP coadd, response, variance, formal-weight, covariance, empirical- noise or significance product. Conditional FRUIT obligations manufacture none. Formal weight is not covariance/significance. Frozen descriptions do not fill absent numerical parameters, support/adequacy or response authority.

3 Composition, fixed-state derivatives and model-error transfer

Bind $\Pi_k : M_k \rightarrow Y_k$, $F_k : Y_k \rightarrow Z_k$, $B_k : M_k \rightarrow Z_k$ with compatible operand pairs, support, reference/gauge/null-space state and all consequential state. The ordinary conditional composition is

$$\rho_k = y_k^{\text{in}} - \Pi_k(m_k^{\text{applied}}), \quad z_k = F_k(\rho_k) + B_k(m_k^{\text{applied}}). \quad (1)$$

$\Pi_k(m_k^{\text{applied}})$ is the removal contribution; $F_k(\rho_k)$ is processed residual; $B_k(m_k^{\text{applied}})$ is rejoin contribution. The exact operation graph specifies order, bypassed operations, removal/rejoin locations and all path application counts. No inverse, $\Pi_k = B_k$, equal spaces/supports, universal upstream bypass, model truth or unit response is assumed. Unknown contributions are not zero.

Only when the *complete* $F_k = A_k$ is exact fixed linear on the declared domain, support and state does distributivity give

$$z_k = A_k y_k^{\text{in}} - A_k[\Pi_k(m_k^{\text{applied}})] + B_k(m_k^{\text{applied}}). \quad (2)$$

Nonlinear Π_k/B_k keep this map-valued form. Only when A_k, Π_k, B_k are all exact fixed linear maps on compatible vector spaces/coordinates may one use

$$D_k := B_k - A_k \circ \Pi_k, \quad z_k = A_k y_k^{\text{in}} + D_k m_k^{\text{applied}}. \quad (3)$$

A fixed basis or input-dependent internal-coordinate evaluation alone proves neither complete linearity nor nonlinearity; all centering, support and material state count. Affine offsets cannot be discarded. Otherwise retain the full composition.

For differentiable fixed numerical maps, the chain rule on exact compatible tangent/reference/gauge spaces gives

$$\begin{aligned} J_{z \leftarrow y, k}^{\text{fixed}} &= DF_k|_{\rho_k}, \\ J_{z \leftarrow m, k}^{\text{fixed}} &= DB_k|_{m_k^{\text{applied}}} - (DF_k|_{\rho_k}) \circ (D\Pi_k|_{m_k^{\text{applied}}}). \end{aligned} \quad (4)$$

Bind baseline, perturbation units/basis, support, tangent domains/codomain, derivative convention and fixed conditioning. Complete fixed linearity reduces these maps to A_k and D_k . Failed differentiability or branch/support/gauge comparison makes the derivative unavailable, or permits a separately typed finite-difference/state-transition object with its own convention. This fixed derivative never substitutes for RF-03–RF-07.

Under complete fixed linearity, the same input/operators and compatible applied reference/error space, subtraction of two results gives

$$\begin{aligned} \Delta z_k^{\text{model}} &= z_k(m_{\text{ref}}^{\text{applied}} + \delta m_k^{\text{applied}}) - z_k(m_{\text{ref}}^{\text{applied}}) = D_k \delta m_k^{\text{applied}}, \\ b_{z, k}^{\text{model}} &= \mathbb{E}_{\mathcal{L}}[D_k \delta m_k^{\text{applied}} \mid \Omega_{k, \text{pre}(\mathcal{L})}^{\text{FRUIT}}] \quad \text{only with exact bias authority.} \end{aligned} \quad (5)$$

Here D_k is the **model-path transfer operator**; $\Delta z_k^{\text{model}}$ is the **model-induced output shift**. These are distinct products, not interchangeable names. Model mismatch is the declared model/reference discrepancy, not covariance or bias by implication. Calling the law-qualified expected shift model-induced bias additionally requires an exact target estimand, reference/truth model, model-error random variable, probability law and its exact law-relative conditioning. Without those, bias is unavailable while the declared shift/sensitivity may remain available. Transform accepted error through its exact acceptance-to-application relation first.

4 Operator-defining state and internal application coordinates

Every use of “coefficient” shall identify its scientific role. Three classes are distinct; numerical changes alone do not classify the response.

A. Operator-defining state and coefficients.

These determine the complete induced input-output map and are fixed in RF-01/02. For the admitted PTC application bind loading subspace $\hat{A}_g$, application metric/influence $W_{g,t}^{\text{app}}$, normal matrix $G_{g,t}^{\text{app}}$ and its generalized inverse, fixed centering/reference λ_g , detector/group binding, application/output support, rank requirement and frozen tolerance, mask, projection orientation/family, branch, generation and every other fact changing the induced map. A fixed structure alone does not fix unresolved defining state.

B. Input-dependent internal application coordinates.

These intermediate solves are functions evaluated from the perturbed operand while A-class facts remain fixed. PTC's $\hat{m}_{g,t}^{\text{app}}(x)$ is such a modal coordinate. Its changed value alone is not operator-defining coefficient relearning, operator re-resolution, a changed method/operator generation or RF-03. It is evaluation of the same map. Preserve its operand/query/value bindings; it is not the FRUIT applied feedback model.

C. Frozen numerical removed component.

A component $\hat{U}$ computed from one parent and then held numerically fixed under perturbation defines the separately named affine subtraction family $\mathcal{A}_{\hat{U}}^{\text{sub}}(Y') = Y' - \hat{U}$ with $\text{D}\mathcal{A}_{\hat{U}}^{\text{sub}}[H] = H$. It is not ordinary fixed-subspace projection and is not the base PTC PCA/SVD application rule.

Relearning or resolving an A-class fact is procedure variation. Evaluating a B-class coordinate is fixed-map application. Holding a C-class component fixed is a different named affine operation. The seven operation-state categories remain; these three roles clarify their contents, not three new lifecycle states. No family, rank, support, method or route is selected, and no response fidelity is established by this conditional classification.

5 Exact admitted PTC fixed-subspace specialization

For one exact frozen group/time application state, substitute the admitted PTC modal-coordinate equation into its output equation. All symbols in this section are PTC-local; $z_{g,t}$ is the upstream operation's output, not by itself the complete FRUIT result z_k .

$$\begin{aligned} G_{g,t}^{\text{app}} &= \hat{A}_g^\top W_{g,t}^{\text{app}} \hat{A}_g, & \hat{m}_{g,t}^{\text{app}}(x) &= x_{g,t} W_{g,t}^{\text{app}} \hat{A}_g (G_{g,t}^{\text{app}})^+, \\ P_{g,t,\Theta}^{\text{PTC}} &= I - W_{g,t}^{\text{app}} \hat{A}_g (G_{g,t}^{\text{app}})^+ \hat{A}_g^\top, & & \\ z_{g,t} &= x_{g,t} - \hat{m}_{g,t}^{\text{app}}(x) \hat{A}_g^\top = x_{g,t} P_{g,t,\Theta}^{\text{PTC}}, & \delta z_{g,t} &= \delta x_{g,t} P_{g,t,\Theta}^{\text{PTC}}. \end{aligned} \tag{6}$$

The identity I acts on the exact detector row domain. Bind all matrix axes and the source's detector-right orientation. Θ denotes all A-class facts, including G and its generalized inverse. The coordinate $\hat{m}^{\text{app}}(x)$ is reevaluated algebraically when x changes; $P_{g,t,\Theta}^{\text{PTC}}$ remains fixed. This establishes the fixed-state input response for that operation, within RF-01 for its declared fixed-map use. A full FRUIT response must compose its actual typed path maps; this local identity does not replace that composition or authorize a parent route.

For the uncentered calibrated input and fixed additive reference:

$$\begin{aligned} x_{g,t} &= Y_{g,t}^{\text{CAL}} - \lambda_g, \\ z_{g,t} &= (Y_{g,t}^{\text{CAL}} - \lambda_g) P_{g,t,\Theta}^{\text{PTC}}, \quad D_Y z[H] = H P_{g,t,\Theta}^{\text{PTC}}. \end{aligned} \tag{7}$$

This map is fixed affine, not strictly linear in Y^{CAL} by implication. The term $-\lambda_g P_{g,t,\Theta}^{\text{PTC}}$ remains part of the output. Its derivative is detector-right multiplication by P , unlike the identity input derivative of frozen-component subtraction. No numerical equality of one output makes those operations interchangeable.

The source admits the solve only when every required input and G^{app} are finite and $\text{rank}_{\tau_g}(G^{\text{app}}) = k_{\text{req},g}$ under exact frozen tolerance, with exact application/output support. The same admitted domain applies to the perturbation. Leaving it yields affected unavailability/cause or a separately typed transition; it does not silently change the RF role, reduce rank, fill data or borrow detectors.

If perturbation reruns centering, subspace, metric, support/mask, detector binding, rank/tolerance, grouping, selection or another defining-state rule, the query is RF-03 or another separately typed procedure response. If exact source/state is insufficient to establish P for a particular case, that case remains unavailable. PTC owns the estimator and its map; FRUIT owns only the response-role assignment to a declared use. No estimator, PTC family, rank/support rule, numerical FRUIT method/route or response fidelity is changed or approved by this source-derived crosswalk.

6 Response conditioning: fixed maps and state-changing procedures

Every response record binds role, perturbation source/domain, units, basis, baseline, codomain, derivative or finite-difference convention, fixed/rerun state, recurrence/selector extent, operation counts, generations and conditioning. Baseline/perturbed support, branch and reference/gauge/null-space comparison shall be explicit. Changed reference is not silently signal. Core declarations supply no numerical availability.

RF-01: fixed one-step measured-input response.

Perturb only $y_k^{\text{in}} \in Y_k$ while freezing the complete realized operator-defining state and induced numerical Π_k, F_k, B_k maps, applied model and every other consequential fixed fact. Evaluate every internal coordinate required by those fixed maps on the perturbed operand; such coordinates may change. Codomain is the compatible Z_k difference/tangent space. No defining-state relearning or re-resolution, model learning, recurrence, stopping or selector rerun belongs to this query. Measured-input names the iteration-input path, not a new independent measurement.

RF-02: fixed one-step applied-model response.

Perturb only $m_k^{\text{applied}} \in M_k^{\text{applied}}$ while freezing the complete realized operator-defining maps/state and measured iteration input. Evaluate every internal fixed-map coordinate required by the applied-model action, including its removal, residual-processing and rejoin paths. Codomain is the compatible Z_k difference/tangent space. Candidate/accepted perturbations must separately include their transformations. No construction/acceptance, defining-state learning or re-resolution, recurrence or selector rerun belongs here.

RF-03: one-iteration full-procedure response.

Perturb the declared one-iteration input with prior S_k and upstream producer procedure/state fixed. Rerun every method-prescribed operation that can change the operator-defining map or consequential state: model construction/acceptance, loading subspace, metric, support/mask, centering/reference, rank, grouping, selection, threshold, branch or other state-resolution rule. Evaluate required internal coordinates within the resulting maps. Bind the result codomain and all state/support/branch transitions; earlier recurrence and terminal selection are outside this local role.

RF-01/02 evaluate fixed maps; RF-03 reruns prescribed defining-state resolution. Equal resulting maps do not erase the different causal query. An internal coordinate change under one fixed map is not RF-03 merely because it is called an application coefficient. A recomputed quantity that changes defining state/map belongs to RF-03 or another separately typed procedure query. The declared fixed map must be established from exact source/state bindings; the admitted PTC specialization supplies that relation on its stated domain. Insufficient source/state evidence gives typed unavailability, not a blanket recomputation rule.

Leaving the exact finite/rank/support domain makes the affected fixed response unavailable or a separately typed transition/discontinuity; it does not silently become RF-03. Fixed derivatives use the general chain rule and reduce to A_k, D_k only under complete linearity. A fixed affine operation retains its offset even when its derivative is linear. Changed method-defining rules need a new method identity; merely evaluating a fixed map does not.

7 Recursive, terminal, whole-chain and parent-domain responses

RF-04: fixed-sequence recursive response.

Perturb the declared recursion input; fix method rules, initial state, upstream producer procedure/state and the predeclared sequence or absolute endpoint index. Propagate input/model/state dependence through that exact fixed extent. Bind recursion-input and result/sequence codomains and query convention. Data-dependent sequence length and terminal selection are outside this role.

RF-05: selected-terminal response.

Perturb the declared procedure/selection input, naming fixed initial/upstream facts. Include consequential iterations, convergence, stopping, resource limits and terminal selection. Bind complete candidate population, conversions, comparison rule, causes, terminal-record identity, selected-result codomain and query convention. Include branch/support/state/selection transitions.

RF-06: whole-chain response.

Perturb the exact upstream/source query, naming fixed external facts. Rerun every included data-dependent upstream producer and the declared FRUIT procedure, including its recurrence/terminal extent. Bind upstream/final domains, producer identities and query convention; unavailable included producer or comparison dependencies remain unavailable.

RF-07: complete FRUIT-only parent-domain response.

Perturb the realized parent-domain payload while holding upstream producer operation, learned state and generation fixed. Compose declared parent-domain entry with the complete FRUIT procedure, including recurrence and terminal selection as applicable. Bind initial FRUIT conditions, parent/final domains, query convention and RF-03/RF-04/RF-05 component references. No upstream producer is rerun. The numerical query payload is not held fixed.

RF-03 is the local one-iteration relation with prior state fixed. RF-04 composes fixed extent; RF-05 includes stopping/selection extent. RF-07 is the outer parent-domain composition and references those components, not an independent alias of any one of them. A one-iteration specialization that reduces to a component reuses its exact response/status with that specialization recorded. RF-06 reruns the included upstream producers; RF-07 does not. Equal numerical values do not erase distinct query identities.

Perturbed parent payloads are separately identified response-query objects with immutable baseline-parent ancestry. Fixed producer generation neither mutates the measured parent nor claims a new upstream realization. All roles retain exact reference/gauge/null-space and branch/support/state/selection comparison. Threshold or selection boundaries may destroy a derivative; undefined comparisons, missing results and nonexistent derivatives retain typed unavailability/discontinuity, with affected scope and cause. A finite difference needs its own declared step, normalization and comparison contract; a fixed-branch derivative is not its replacement. No online adaptive filter or other numerical estimator is introduced.

Scoped supersession. The former RF-07 held-numerical-parent meaning remains retired. r0.3 replaces r0.2's blanket treatment of internal coordinates as frozen defining state, and its generic recomputation-to-RF-03 assignment, with the three exact roles and causal boundaries above. All seven stable RF identities remain.

8 Law-relative stochastic targets and complete joint covariance

Every target binds `target_law_identity`, random variables, conditioned fixed objects, resolution/selection condition, omitted variation, support/domain and reference/gauge/null-space state. $\Omega_{k,\text{pre}(\mathcal{L})}^{\text{FRUIT}}$ means fixed *before the random event of that exact law $\mathcal{L}$* , not before every physical or computational outcome. Bind quantity/units, parent/population/generation, maps/rules, state, grouping and assumptions explicitly; a generation label is no substitute for conditioning. Three distinct targets are required:

Physical/random-parent target.

The measured parent and every model whose variation is estimated remain random. Do not condition them on realized numerical values. Preserve all model-learning variation and parent/model dependence included by this exact law.

Fixed-realized-model target.

The exact applied model is deliberately conditioned fixed under a separately named law/target. Its model covariance and parent/model cross blocks vanish only there; model-learning variation is omitted. This is not equivalent to the random-model target. The stochastic parent is not silently fixed too.

Conditional NOI-randomization target.

The realized parent payload, fixed applied model/operator graph, support and other exact state may be conditioning facts because they precede the declared NOI assignment/member draw. When its source law conditions on successful design resolution, retain that exact resolution condition explicitly. This is neither a physical-parent target nor permission to select favorable FRUIT outcomes.

A branch-, completion-, convergence- or terminal-selection-conditioned FRUIT target requires a separate method and estimand; it remains unavailable without separate authority. Do not hide such censoring or a draw-dependent domain in $\Omega_{k,\text{pre}(\mathcal{L})}^{\text{FRUIT}}$. Record any permitted selection/domain conditioning in its exact source law and target; no universal physical-parent interpretation is imposed on NOI.

Under complete fixed linearity, compatible real vector coordinates, finite joint second moments and exact law/conditioning, define $C_{yy}^{\text{FRUIT}}, C_{mm}^{\text{FRUIT}}, C_{ym}^{\text{FRUIT}}, C_{my}^{\text{FRUIT}}$ as the conditional covariance blocks of $(y_k^{\text{in}}, m_k^{\text{applied}})$ under $\text{Cov}_{\mathcal{L}}[\cdot \mid \Omega_{k,\text{pre}(\mathcal{L})}^{\text{FRUIT}}]$. Axes bind quantity, units, support and reference/gauge/null-space coordinates; m is applied model.

$$\begin{aligned} C_z^{\text{FRUIT}} &= A_k C_{yy}^{\text{FRUIT}} A_k^{\top} + D_k C_{mm}^{\text{FRUIT}} D_k^{\top} \\ &\quad + A_k C_{ym}^{\text{FRUIT}} D_k^{\top} + D_k C_{my}^{\text{FRUIT}} A_k^{\top}. \end{aligned} \tag{8}$$

For a real joint covariance $C_{my}^{\text{FRUIT}} = (C_{ym}^{\text{FRUIT}})^{\top}$. Both cross terms remain unless the exact law/conditioning justifies omission. A parent-learned model is not independent by implication. Fixed gauge or a singular covariance does not mean an unavailable absolute mode was measured with zero uncertainty. Nonlinear/random-operator propagation or a Jacobian approximation requires its own typed method; it is not this exact identity. Accepted uncertainty reaches applied space only through the exact transformation and dependence. Repeated shared-parent iterations are not independent observations and imply no inverse-square-root iteration-count uncertainty law.

9 NOI target-law members, full-product action and source binding

For complete fixed linearity and explicitly fixed applied model/operators/state, full parent realizations and zero-target-law NOI members have distinct actions:

$$\begin{aligned}
 z_{k,b}^{\text{full}} &= A_k y_{k,b}^{\text{full}} + D_k m_k^{\text{applied}}, \\
 \mathbb{E}_{\mathcal{L}}[z_{k,b}^{\text{full}} \mid \Omega_{k,\text{pre}(\mathcal{L})}^{\text{FRUIT}}] &= A_k \mathbb{E}_{\mathcal{L}}[y_{k,b}^{\text{full}} \mid \Omega_{k,\text{pre}(\mathcal{L})}^{\text{FRUIT}}] + D_k m_k^{\text{applied}}, \\
 \mathbb{E}_{\mathcal{D}_{k,g}^{\text{NOI}}}[\delta y_{k,b}^{\text{NOI}} \mid \mathcal{H}_{k,g}^{\text{NOI}}] &= 0, \quad \delta z_{k,b}^{\text{NOI}} = A_k \delta y_{k,b}^{\text{NOI}}.
 \end{aligned} \tag{9}$$

Here k is absolute iteration, b the member, and g the ensemble generation. Bind each object's full versus NOI role, exact parent/output domain, units, support/reference/gauge/null-space, target law and center convention. $\mathcal{L}$ in the full-product expectation is its separately declared law; $\mathcal{D}_{k,g}^{\text{NOI}}$ is the exact conditional NOI design/assignment law and $\mathcal{H}_{k,g}^{\text{NOI}}$ its fixed-parent/model/operator/support/design-resolution conditioning. These are role-qualified symbols, not PTC transformed signal or the result-space Z_k .

The zero is a **target-law expectation**. No member is required to have zero sample or spatial mean. No finite-ensemble mean subtraction, member-wise centering or $B - 1$ correction is implied. Source and deterministic residual structure may remain in individual members. The deterministic $D_k m_k^{\text{applied}}$ term shall not be added to each zero-target-law member. It belongs to the full product and its exact law-qualified center.

The exact content-bound owner source is `OWNER_INTERIM_CORRECTION_R0.2.txt`, section 2, with its directive identity/digest in the source record and preserved r0.2 archive bound on each cover. It authorizes this conditional zero-target-law distinction; it does not supply a complete external numerical NOI law or estimator. An actual source-law identity/digest, random event, conditioning/design resolution, ensemble generation, exact NOI-owned product and estimator binding remain `unavailable_external_binding` until supplied through separately approved authority. A symbolic $\mathcal{D}_{k,g}^{\text{NOI}}$ is not an admitted law instance. Where an exact law supplies known-zero second moments, use its quantity-appropriate zero-target-law action and estimator convention. No numerical FRUIT/NOI route follows from the equation.

Every boundary names member meaning: full parent realization, zero-target-law NOI member, residual, complete FRUIT output or another exact quantity. A full-affine-member method requires exact member/transformed centers, source-law and estimator authority, centering/second-moment convention, response/support parity and a separately bound method. No finite-ensemble centering is inferred.

Fixed-state NOI requires exact application, response and support parity and explicit omitted learning/support/selection/stopping/restart/leakage variation. NOI-informed later learning creates a dependent immutable science/ensemble/ uncertainty successor, not independent validation or retroactive editing. Per-realization relearning of model/support/cleaner/operator-defining coefficients/stopping/selector is a distinct method and ensemble with complete member-state graph. New values under unchanged rules produce new states, not a method each. No mixing/pooling of fixed and relearned members is authorized here. FLT and other parent routes retain exact producer quantities and response authority; no filter inverse, route permission or interchangeability is supplied by FRUIT/NOI naming.

10 Closed uncertainty-role grammar and law-bound availability

The closed FRUIT uncertainty-role grammar has prefix **SCI-FRUIT/UNCERTAINTY/** followed by exactly one terminal token in this closed table. Concatenating prefix and token is the exact product identity; no undeclared **<role>** placeholder is admitted.

MEASURED-PARENT-COVARIANCE

Exact random-parent law and compatible covariance axes.

CANDIDATE-MODEL

Candidate-model uncertainty, construction and any candidate prior; not accepted covariance.

CANDIDATE-SELECTION

Selection dependence and its effect on the exact target.

ACCEPTED-MODEL-COVARIANCE

Accepted axes/support/reference; not copied into applied space.

ACCEPTANCE-TO-APPLICATION

Exact transformation uncertainty and dependence, including deterministic facts.

APPLIED-MODEL-COVARIANCE

Applied axes and law used by conditional propagation.

PARENT-APPLIED-CROSS-COVARIANCE

Both compatible orientations and dependence; no independence by omission.

OPERATOR-STATE

Consequential operator/state variation; exact method required.

SUPPORT-THRESHOLD-SELECTION

Exact support/threshold-selection law and domain.

STOPPING-TERMINAL-SELECTION

Exact rule/population/selection-conditioned target authority.

EXTERNAL-PRIOR

External prior law, origin and dependence, separately from model-stage products.

EMPIRICAL-REPEATABILITY

Exact repeat population; no automatic analytic covariance or coverage.

NOI-PRODUCT-REFERENCE

Immutable reference to the exact NOI-owned uncertainty product identity, generation and content digest. The external role/law/estimator binding is currently **unavailable_external_binding**; this FRUIT reference does not invent or replace the NOI-owned product.

Every role independently binds quantity/domain/axes, units (including covariance product units), target law/random variables/fixed objects/resolution condition, omitted variation, support/reference/gauge/null-space, availability/cause and generation. One storage container is allowed only when all distinctions remain lossless. Accepted covariance reaches applied covariance only through exact transformations and dependence, including transformation uncertainty.

Model mismatch, model-path transfer, model-induced output shift and model-induced bias have the distinct product identities above. They are not implicitly members of this covariance/uncertainty family. A deterministic discrepancy or shift is not covariance; a computable shift does not establish bias. No absent source, mode or estimator becomes measured zero uncertainty.

11 Path support, occurrence classes and information origin

All eight roles are required and independently addressable; no method may waive them. Bind exact scientific queries and path/operation/application identities, with materialized or exactly reconstructible values or permitted typed status.

1. **Measured-parent support:** original measured occurrences, producer availability, domain and causes.
2. **Model-definition support:** separate candidate/accepted/applied spaces and definition supports, parent/prior origin, transformations and any exact equality.
3. **Projection/removal support:** where the projected applied operand and compatible subtraction exist, including mapping/boundary/missing-data behavior.
4. **Residual-processing support:** every operation's input/output support, retained membership and causes.
5. **Model-rejoin support:** where the exact rejoin contribution exists, independently of removal support.
6. **Final output support:** the declared output domain and permitted composition.
7. **Learning influence:** transitive dependence of learned/selected state on inputs, not local membership or acquired exposure.
8. **Response-query support:** exact role, source basis/domain, conditioning, output support and limitations.

A common mask, array shape or coordinate label does not establish identity. Acceptance does not imply application on identical support. Each operand pair still requires full quantity/reference/gauge/null-space compatibility.

`residual_and_model_supported` means both contributions exist and may be combined under the exact method. `residual_only` and `model_only` require explicit permission for the narrower path composition; absence is not a numerical zero. `unavailable` retains cause and affected scope when the requested output lacks a permitted composition or required input. A method may prohibit any class. Merely having a model at an unmeasured location does not authorize model-only output.

Undefined unequal-support behavior is unavailable. No implicit zero fill, extrapolation, imputation, renormalization, nearest substitution, model completion, reference offset or gauge alignment is permitted. Any mapping/restriction into a common result space shall be exactly bound by the method.

An upstream unavailable absolute/additive mode remains observationally unavailable after rejoin. Its contribution is `model_or_prior_supported`, not `observationally_measured`, `recovered_from_data`, `acquired_exposure` or `unit_response_truth`. This is a mode-origin state, not a fifth occurrence class; even a residual-and-model occurrence may contain it. Record reference/gauge/null-space identity and response/uncertainty consequences. Choosing a gauge creates no observation or measured zero uncertainty. Iteration, projection and rejoin create no exposure or independent measurement. Original producer exposure, path contribution and transitive learning influence remain distinct; no absent exposure product is invented. Shared parents/models/calibration/response/reference preserve dependence.

12 Exact product identities and scientific handoff

Every instance binds exact role, method, immutable parents, quantity/domain/units, support/origin, state/application/iteration/branch/lineage/product generation, availability, outcome/cause and provenance. The core-declared product roles are:

SCI-FRUIT/ITERATION-RESULT

Child numerical payload z_k . Scientific consumption requires the exact ITERATION-BUNDLE or its immutable exact reference; a bare array is not a complete handoff.

SCI-FRUIT/ITERATION-BUNDLE

All universal and method-required roles and exact bindings.

SCI-FRUIT/SUCCESSOR-STATE

Produced S_{k+1} and causal lineage; sufficiency remains distinct.

SCI-FRUIT/CONTINUATION-STATE

Exact future-consequential state queries, reconstruction and sufficiency status.

SCI-FRUIT/PATH-SUPPORT

Eight separately bound support/influence roles.

SCI-FRUIT/INFORMATION-ORIGIN

Occurrence/path origin and separate mode-origin state.

SCI-FRUIT/TERMINAL-SELECTION

Separate immutable selector over completed iterations.

SCI-FRUIT/TERMINAL-PRODUCT

Selector plus immutable selected completed bundle or exact reference.

SCI-FRUIT/MODEL-PATH-TRANSFER

D_k under complete fixed linearity; operator domain/codomain, units and conditioning are exact.

SCI-FRUIT/MODEL-INDUCED-OUTPUT-SHIFT

$\Delta z_k^{\text{model}} = D_k \delta m_k^{\text{applied}}$; deterministic difference is not covariance or bias.

SCI-FRUIT/MODEL-INDUCED-BIAS

$b_{z,k}^{\text{model}}$ only with exact estimand/reference/error-law/conditioning authority; may be unavailable while transfer/shift exist.

SCI-FRUIT/MODEL-MISMATCH

Exact model/reference discrepancy and domain; separate from bias, output shift and covariance.

The core-declared response-role vocabulary is SCI-FRUIT/RESPONSE/RF-01, SCI-FRUIT/RESPONSE/RF-02, SCI-FRUIT/RESPONSE/RF-03, SCI-FRUIT/RESPONSE/RF-04, SCI-FRUIT/RESPONSE/RF-05, SCI-FRUIT/RESPONSE/RF-06, and SCI-FRUIT/RESPONSE/RF-07. Each binds the corresponding exact query, conditioning, domains, state/generation and status defined in the response sections. This is a closed declaration of these seven roles, not an unexplained placeholder or a claim that a numerical response exists.

These roles create no new generic numerical parent class. Scientific consumption also retains the exact underlying MAP, JINC, FLT-FIXED, FLT-MATCHED, detector-time or other method-defined producer quantity and its admission authority.

13 Immutable request state and producer-stage branching

request_state is **not_requested** or **requested** for one immutable request generation. There is no transition from not-requested to requested within it; a later request creates a new request generation. For a requested generation the successful producer-stage order is

$$\begin{aligned} & \text{requested} \rightarrow \text{effective} \rightarrow \text{application_scope_resolved} \rightarrow \text{applied} \\ & \rightarrow \text{iteration_realized} \rightarrow \text{complete_iteration_candidate} \rightarrow \text{completion_decided} \\ & \rightarrow \text{completed_iteration} \rightarrow \text{published.} \end{aligned}$$

requested binds the request; **effective** the effective action; **application_scope_resolved** its exact resolved scope/state. **applied** means the operation was invoked. **iteration_realized** means immutable numerical path values and execution-outcome records exist. **complete_iteration_candidate** means every required role has an exact value or a core-and-method-permitted status. **completion_decided** records the FRUIT-owned exact local decision. **completed_iteration** means that candidate satisfies the rule. **published** exposes the immutable completed product under its role.

Separately type scientific availability as **available** or **unavailable**, and producer outcome as **realized**, **failed** or **not_produced**. Every branch/role outcome binds stage actually reached, affected scope, cause and generation. These are not mandatory steps in the successful path. Early stop shall not fabricate applied, realized, completion or publication records. Preserve every stage actually reached and every attempt. Unavailable optional companions may coexist with a narrower complete product only when both core and exact method permit it.

superseded is an immutable replacement relation, not erasure or a mandatory producer stage. Restart/retry/new-request identities preserve their parents and cannot relabel a stopped attempt as success. Terminal selection consumes completed iterations and creates a separate immutable selector record; it does not mutate the selected bundle.

Downstream decisions are separate child records. Base iteration and terminal completion do not depend on absent or failed downstream use. An exact explicitly requested FRUIT product role may require a child; that dependency is local to that declared requested role, never an implicit requirement of the base roles.

14 Operation state, method identity and exact continuation

S_0 is the initial seed. The generic transition records roles without selecting an update or recurrence:

$$(S_k, y_k^{\text{in}}) \longrightarrow (m_k^{\text{candidate}}, m_k^{\text{accepted}}, m_k^{\text{applied}}, \rho_k, z_k, S_{k+1}).$$

Every method declares whether y_k^{in} is original parent, previous result, reconstruction from base parent/cumulative model or another exact product, with every domain conversion. Returning to base and reprocessing a prior result are different methods even with the same update. Immutable measured base ancestry persists. Causal update, cumulative model, diagnostic difference, accepted/applied model and iteration result are separate roles; no increment meaning is inferred from z_k .

A method-rule change in parent route, grouping, recomputation, learning/model construction, operator family, sign, order, bypass graph, support, recurrence, stopping, terminal selector or failure behavior requires a new method identity. Changed operator-defining coefficients, learned state or model values under unchanged resolution rules require distinct realized state/application generations, not a new method each. Internal application coordinates evaluated by the same fixed map may change without relearning, re-resolution, a new method or a new operator generation. Their operand/query/value provenance remains explicit. Use grouping-neutral `application_scope_resolved` and distinct `learned_candidate`, `resolved_model_state`, `applied_model_state`, `realized_iteration_state`; not every model is fitted. Requested, effective, resolved and realized remain distinct.

Factor each operation and consequential component into exactly one category at its exact conditioning scope: **fixed numerical operator** (exact map, state; linearity separate); **fixed structural state with recomputed operator-defining coefficients** (structure, resolution rule, dependencies, realized map); **externally supplied** (owner/parent/version, not implied independent); **relearned** (rule, population, dependence, generation, failure); **bypassed** (exact absent model-path operation, locations/counts on both paths); **not_applicable** (permitted absent role/reason); **unavailable** (missing/incompatible fact/cause/domain/blocked claim). Mixed stages are factored; each entry binds owner, parent/generation, inputs, order, count, path and failure. Different conditioning scopes use separate records. Internal fixed-map coordinates are evaluation values, not an eighth operation-state category; their changes alone do not convert a fixed operator into a re-resolved one.

Every method shall bind its exact stochastic generator/algorithm, keys/counters/ state, populations and sampling controls where applicable; otherwise it shall record a justified **not_applicable**. No numerical values are selected here.

After completion of the iteration with absolute zero-based index N , continuation begins at absolute index $N + 1$ from S_{N+1} . N is an index, not a count. Exact continuation requires the same compatible future controls/ numerical profile and all state capable of changing future required results: base parent; candidate/accepted/applied and cumulative models; learned/operator state; supports/masks; update/selection/convergence history; counters/resources; stochastic algorithm/generator/key/counter state where applicable; method/ parameter versions; branch/generations; reference/gauge/null-space state and all upstream/model parents. No finite list excuses another consequential fact. Exact state queries/reconstruction/failure contracts may be materialized, compact or lineage-reconstructible. Representation alone does not change state. Continuation must agree in scientific state/results with uninterrupted execution.

Restart creates a new lineage with explicit reused ancestry; initialization/ counters follow its method. Relearning under unchanged rules may be an exact continuation successor. Branch records departure and parent state. Retry is a new attempt with exact state/randomness/failure accounting, not a new observation or implicit reset. Seed, continuation, restart, successor, branch, retry, completion, terminal selection and publication retain separate identities.

15 Completion, comparable terminal candidates and downstream decisions

Universal iteration-bundle roles are: immutable base parent; exact input; method/lineage/branch/absolute iteration; target and separate candidate/accepted/ applied models/spaces/transformations/equality facts; projected removal, residual, processed residual, rejoin and result; update/successor; all path support/influence and information origin; response/uncertainty states; validity/ causes; continuation state; completion/failure and provenance. Removal/rejoin bind full additive/reference/gauge/null-space compatibility and unavailable-mode consequences. Path values are available or exactly reconstructible on claimed support.

No method may waive universal roles or hard dependencies. A numerical result needs exact input/applied operand and all required compatible contributions; continuation needs all causal state/profile compatibility; a numerical response/ uncertainty needs its exact premises/inputs/domains/conditioning; terminal selection needs a separate immutable reference to a completed compliant bundle. An absent role requires core-permitted **not_applicable** plus the method's narrower composition. Unknown is unavailable, never zero, identity, bypass or omission. Unavailable companions permit only narrower claims allowed by both core and method. Numerical draft completion is not claimed here.

Keep iteration completion, resource limit, convergence, policy stop, failure, terminal selection and consumer fitness separate. Convergence binds proposition, quantity/domain, metric/norm, scale/denominator, tolerance, direction, persistence/consecutive rule, missing/non-finite behavior, generation and state. No value is chosen. Limits/completion do not prove convergence; convergence proves neither correctness, unit response, unbiasedness nor fitness.

A terminal selector consumes one or more completed iterations using one exact predeclared rule with data access, ties and failure behavior. Candidates shall share, or be mapped by exact selector-owned transformations into, common quantity/role, unit/scale, frame/grid/domain, additive reference/gauge, response meaning, support comparison domain, method/generation relation and uncertainty/metric meaning. Shape or a common scalar diagnostic is insufficient. Comparing methods/routes/groupings/gauges/generations/supports makes every conversion/comparison rule part of selector method identity. Unavailable comparisons, causes and exclusions remain, never a default last/latest, lowest-residual or visually favorable choice.

Nine terminal fields remain required: selected completed identity; full candidate population with every inclusion/exclusion and conversion; exact rule/generation and ties/failure; convergence/stop/resource facts; selection causes; RF-05 response/status with any RF-07 composition; exact terminal uncertainty/conditioning/ omissions; continuation/terminality state; requested linked named-use decision identities and exact statuses, or an explicit statement that no downstream use was requested. All retain scope, generation, reference/gauge/null-space consequences and cause. Missing a hard selection dependency blocks selection. The immutable selector never mutates, relabels or replaces the selected bundle.

16 Open-world named use and SCI-VAL accounting

Field 9 is open-world: retain only explicitly requested or linked downstream records, not an exhaustive list of future uses. For SCI-VAL-governed use retain separate exact request, applicability, eligibility and realization states plus profile/source binding. Not-requested creates no eligibility proposition. Missing profile/unresolved scope may yield a realized accounting record with **applicability_unknown** and **decision_unavailable**; generic **unavailable** is not a SCI-VAL realization value. Incomplete, failed or not-produced decision artifacts are separate realization conditions. Exact external SCI-VAL profile/source bindings remain unavailable when not supplied; the owner-directed distinctions above do not invent them.

Pointing/OOF/source/catalogue/production admission remains separate child work. Absent or failed downstream use does not erase base local completeness. Only an exact explicitly requested FRUIT product role can require a specified child; that narrower dependency does not alter base iteration/terminal roles. Later child records reference immutable parents without mutation or a new FRUIT iteration generation. SCI-VAL evaluates an exact approved use policy; it does not run FRUIT, advance recurrence, select the model or choose the terminal.

17 Assumptions, required method record and claim ceilings

SCI-FRUIT-ASM-001: compatible arithmetic.

Every additive equality/difference/cancellation uses exact pairwise quantity, unit/scale, frame/domain, grid/sampling, support, origin/reference/gauge, null-space, calibration and response compatibility, including declared affine/quotient arithmetic and state. Missing premises block affected claims.

SCI-FRUIT-ASM-002: fixed residual linearity.

The map-valued distributive expansion requires complete $F_k = A_k$ exact fixed linear on declared spaces/support/state. Fixed basis or recomputation alone is insufficient; affine offsets remain unless the premise actually holds.

SCI-FRUIT-ASM-003: complete fixed linearity.

A_k, Π_k, B_k are all exact fixed linear on compatible spaces, units, support, state and conditioning before defining D_k , its linear applied-model response or the exact four-term covariance. General fixed derivatives require their own explicit differentiability/tangent/comparability premises, not this assumption.

SCI-FRUIT-ASM-004: joint conditional covariance.

In addition to ASM-003, compatible real vector coordinates, finite joint second moments and exact law-relative $\Omega_{k, \text{pre}(\mathcal{L})}^{\text{FRUIT}}$ are required, with applied-model axes, reference/gauge/null-space state and both cross terms. Physical/random-parent, explicit fixed-realized-model and conditional NOI randomization laws are distinct. Pre-outcome is relative to each declared random event; NOI may condition on realized parent and exact successful design resolution. FRUIT outcome-conditioned targets need separate method/estimand authority.

SCI-FRUIT-ASM-005: response comparison.

Bind exact numerical-fixed versus prescribed-rerun query, domains/codomain, state/support/branch/reference/gauge comparison and convention. A derivative requires existence on compatible tangent spaces. A finite difference has a separate convention and is not a substituted fixed-branch derivative.

SCI-FRUIT-ASM-006: causal continuation.

All future-consequential state, compatible profile and the same future controls are required for equality with uninterrupted execution. A selected array alone is insufficient; no replay or numerical sufficiency is established by this core.

Assumptions are conditional premises, not numerical evidence.

Every method-specific record shall satisfy all twenty-one complete binding classes in REQ-023 and their exact obligations in the scientific sections. Optional absence needs core-and-method permission and a reason. A filled template supplies neither actual authority nor numerical availability.

No numerical method/route, representation/response/covariance fidelity, recovered measured mode, uncertainty coverage, convergence adequacy, Pointing/OOF/catalogue fitness, observational validation, performance, readiness, production suitability/ authorization or Unity activity is claimed. Implementation is not assessed; numerical conformance/experiments are not performed. Numerical development, qualification and policy recommendation retain separate owner decisions and independent-pointing evidence obligations; no evidence plan is supplied here.

18 Normative requirements 1–5

SCI-FRUIT-REQ-001: Authority separation

The scientific authority record shall distinguish approved program-order authority A, approved conditional-composition authority B and unavailable numerical-method authority C. Neither authorship, verification, consistency review nor conditional freeze shall imply numerical approval. Every numerical instance shall have an exact FRUIT-FEEDBACK-METHOD/<method-name>@<revision> identity, complete scientific record, separate Grant Wilson approval and required upstream authority. Missing approval shall retain `unavailable_pending_separate_owner_approval` for every dependent numerical route and result.

SCI-FRUIT-REQ-002: Typed objects and spaces

Every scientific object shall bind quantity, units/scale, frame/domain, grid/sampling, shape/axes/indexing, timing where relevant, grouping, support, validity/missing/non-finite causes, parent/producer and generations. The input, residual, processed residual, projected removal, rejoin, result, target, state and each model role shall be separately typed. Shape, coordinate, filename or label agreement shall not substitute for scientific identity. All declared map domains and codomains shall be compatible with their actual operands and output role.

SCI-FRUIT-REQ-003: Candidate, accepted and applied models

Candidate, accepted and applied models shall have separately bound objects, spaces, supports, origin/prior, construction/selection/acceptance/application rules and generations. Every intervening transformation and its allowed inputs shall be exact, including normalization, clipping, sign, projection, restriction and unit conversion where used. Equality of roles or any property requires an exact method/state fact. Removal and rejoin shall consume the applied model. Acceptance grants feedback-model state only, not truth, detection, posterior or downstream fitness.

SCI-FRUIT-REQ-004: Additive compatibility

Removal and rejoin shall each bind compatible quantity, units/scale, frame/domain, grid/sampling, support, additive origin/reference or gauge, null-space state, calibration and response conventions. Spaces shall be exact additive spaces or explicitly declared affine/quotient spaces with well-defined coordinate/quotient arithmetic and one reference/gauge per operation. Every alignment/conversion shall be explicit. Incompatible or missing facts shall make the affected arithmetic and dependent claim unavailable; arbitrary representatives, implicit offsets and inferred equality of input/output spaces are forbidden.

SCI-FRUIT-REQ-005: Exact conditional composition

A method shall bind the exact $\Pi_k : M_k^{\text{applied}} \rightarrow Y_k$, $F_k : Y_k \rightarrow Z_k$ and $B_k : M_k^{\text{applied}} \rightarrow Z_k$, sign, support/state and operation graph for $\rho_k = y_k^{\text{in}} - \Pi_k(m_k^{\text{applied}})$ and $z_k = F_k(\rho_k) + B_k(m_k^{\text{applied}})$. Each path contribution shall be available or exactly reconstructible on claimed support. Equal projection/rejoin, inverse maps, equal domains, universal bypass or unknown-as-zero shall not be assumed. Missing required contributions shall block the affected numerical composition.

19 Normative requirements 6–10

SCI-FRUIT-REQ-006: Linearity hierarchy and derivative premises

Distributive expansion shall require the complete residual procedure $F_k = A_k$ fixed linear on the exact domain/support/state. Defining the linear model-path operator $D_k = B_k - A_k \circ \Pi_k$ shall additionally require fixed linear projection and rejoin on compatible spaces. Affine offsets shall remain. Otherwise the full composition shall be used. General fixed-state derivatives shall require differentiability on exact compatible tangent/reference/gauge spaces with bound baseline, units/basis, support, codomain, convention and fixed conditioning; complete fixed linearity then reduces them to A_k, D_k .

SCI-FRUIT-REQ-007: Parent route, grouping and iteration input

Every iteration shall retain immutable original measured-parent ancestry and bind one exact available producer route and complete grouping, membership, axes and scope. The route shall supply quantity, units, frame, response, beam/template, calibration, support, validity, lifecycle and provenance. The method shall declare the iteration-input rule and all domain conversions. Original-parent, prior-result and reconstruction routes are distinct methods; no recurrence follows from the word iteration. Shared parent/model/calibration/reference dependence shall remain explicit; grouping/coaddition shall not imply commutation.

SCI-FRUIT-REQ-008: Eight support and influence roles

Every iteration shall expose separately addressable measured-parent support, model-definition support for each model stage, projection/removal support, residual-processing support per operation, model-rejoin support, final-output support, learning influence and response-query support. Each shall bind its exact scientific query, domain, operation/path/application, parents/generation, value or permitted typed status and cause. A common mask or shape shall not substitute for distinct roles. No method shall waive a role; exact reconstruction is permitted only with a bound reconstruction and failure contract.

SCI-FRUIT-REQ-009: Occurrence classes and mode origin

Composition shall distinguish residual-and-model-supported, residual-only, model-only and unavailable occurrences under exact method permissions and pairwise compatibility. Absent contributions shall not become numerical zero. Undefined unequal-support behavior shall be unavailable; no implicit fill, extrapolation, renormalization or gauge repair is permitted. Model/prior content in an unavailable measured mode shall retain model-or-prior-supported origin, separately from occurrence class, with response/uncertainty consequences. Iteration, learning, projection and rejoin shall create neither acquired exposure nor independent observations nor recovered measured modes.

SCI-FRUIT-REQ-010: Method rules and realized state

Changes to parent route/grouping, learning/construction or operator-defining resolution rule, operator family, sign/order/bypass graph, support, recurrence, stopping, terminal selector or failure behavior shall create a new method identity. Changed operator-defining coefficients, models or state under unchanged resolution rules shall have distinct state/iteration/application generations; internal fixed-map coordinates may change without a new operator generation. Method, state, absolute iteration, application, attempt, branch, lineage and product generations shall remain distinct. Requested/effective/resolved/realized facts shall not be conflated; representation changes alone shall not imply changed scientific state.

20 Normative requirements 11–14

SCI-FRUIT-REQ-011: Operation graph and operator-defining state

Every named operation and consequential component shall bind owner, parents/generations, inputs, ordered location, path, count and failure scope. Retain all seven operation-state categories at exact conditioning scopes. Distinguish operator-defining state/coefficients, internal fixed-map application coordinates and a frozen removed component. Fixed structural state with recomputed operator-defining coefficients shall bind its resolution rule, dependencies, schedule and realized map; relearning binds population/triggers. A changed internal coordinate alone shall imply no relearning, re-resolution, method change, operator generation or RF-03. Factor mixed stages; bind bypass locations/counts on both paths. No fixed-state label alone proves linearity.

SCI-FRUIT-REQ-012: Exact fixed-map local response queries

RF-01 shall perturb only iteration input while freezing complete realized operator-defining state and induced numerical Π_k, F_k, B_k maps, applied model and other fixed facts. RF-02 shall perturb only the applied model under the same fixed-map condition with measured input fixed. Each shall evaluate required internal coordinates of those maps on its perturbed operand; their changed values alone do not change the map or RF role. Bind source/tangent domains, units/basis, baseline, codomain and convention. The admitted PTC specialization has fixed $P_{g,t,\Theta}^{\text{PTC}}$ and derivative $HP_{g,t,\Theta}^{\text{PTC}}$ on its exact domain, retaining fixed affine centering. Insufficient source/state or finite/rank/support-domain failure yields typed unavailability or a separately typed transition, never silent RF-03 identity.

SCI-FRUIT-REQ-013: Procedure and composed response scopes

RF-03 shall rerun all unchanged-method operations that can change operator-defining map or consequential state within one iteration: model construction/acceptance, loading, metric, support/mask, centering/reference, rank, grouping, selection, threshold, branch or other state resolution. Prior iteration state and upstream procedure/state remain fixed. Evaluation of a fixed-map internal coordinate alone is not such a rerun. RF-04 propagates exact fixed-sequence/index dependence; RF-05 includes stopping/resources and terminal selection; RF-06 reruns included upstream producers; RF-07 holds upstream procedure/state/generation fixed and composes the parent-domain entry and FRUIT-only extent with RF-03/04/05 references. Bind causal scope, domains, generations and comparison. Fixed derivatives shall not substitute for procedure queries; unavailable comparisons retain typed causes/transitions.

SCI-FRUIT-REQ-014: Law-relative stochastic targets

Every stochastic target shall bind exact law identity, random variables, conditioned fixed objects, resolution/selection condition, omitted variation, support/domain and reference/gauge/null-space state. Pre-outcome shall mean before that law's random event. Physical/random-parent targets shall retain random parent/model values; explicitly fixed-realized-model targets omit model-learning variation and have zero model/cross blocks only there. Conditional NOI-randomization targets may fix realized parent/model/operator/support and exact source-law design-resolution conditions; they are not physical-parent targets. FRUIT branch/completion/convergence/terminal-conditioned targets need separate method and estimand authority. Under complete fixed linearity and finite compatible joint moments, all four law-qualified covariance terms shall remain unless exact law/conditioning authorizes omission.

21 Normative requirements 15–18

SCI-FRUIT-REQ-015: Core-declared uncertainty, mismatch, transfer, shift and bias

All core-declared uncertainty roles shall remain separately addressable with exact law, random/fixed/resolution facts, domain, units, availability/cause, generation and omissions. Candidate, selection, accepted, transformation, applied and parent/applied cross roles shall remain distinct from other core-declared sources; exact NOI-owned product binding shall not be invented. Accepted covariance/error shall reach applied space only through exact transformation and dependence. D_k shall be MODEL-PATH-TRANSFER; $D_k \delta m_k^{\text{applied}}$ MODEL-INDUCED-OUTPUT-SHIFT. MODEL-MISMATCH and MODEL-INDUCED-BIAS shall have separate identities outside the uncertainty family. Bias requires estimand, reference/truth model, error random variable, law and conditioning; otherwise it is unavailable even when transfer/shift exist. Deterministic difference is not covariance.

SCI-FRUIT-REQ-016: NOI law, member action and source parity

Every FRUIT-to-NOI member shall bind iteration, member, target law, parent/output domains, ensemble generation, center convention and exact source authority or typed unavailable external binding. Under complete fixed linearity with fixed applied model/state, full parents shall produce $A_k y_{k,b}^{\text{full}} + D_k m_k^{\text{applied}}$ and their exact law-qualified center; zero-target-law members shall produce $A_k \delta y_{k,b}^{\text{NOI}}$ without rejoin. Zero means expectation under the exact conditional NOI law, not member/spatial/finite-ensemble mean subtraction or $B - 1$ correction; individual source/residual structure may remain. Full-affine methods need separate center/estimator authority, source-law, response/support parity and method identity. Fixed-state, dependent-successor and relearned methods/ensembles shall remain distinct and admit no numerical route here.

SCI-FRUIT-REQ-017: Continuation and attempt lineage

After completion of absolute zero-based iteration N , exact continuation shall start at $N + 1$ from S_{N+1} and agree with uninterrupted execution under the same future controls and compatible method/numerical profile. It shall preserve every future-consequential scientific/stochastic/control fact or exact reconstruction with failure behavior; a result array alone is insufficient. Restart shall create a new lineage with reused ancestry explicit. Branch, successor and retry shall retain exact parent state, attempts, randomness and failure accounting. No reset, independent observation or sufficiency follows merely from saved state or a successful replay process exit.

SCI-FRUIT-REQ-018: Product handoff and branched lifecycle

Each immutable request generation shall have request_state not_requested or requested; a later request creates a new generation, not an in-place transition. For requested generations, preserve the exact successful producer-stage order and distinct effective/resolved/applied/realized/candidate/decision/completed/published meanings. Scientific availability and producer outcome shall be separately typed at the stage actually reached, with cause/scope/generation; early stop shall fabricate no later invocation, realization, completion or publication. Supersession shall preserve prior records. Exact product declarations, parents and bundle bindings are mandatory: ITERATION-RESULT is a child payload whose scientific consumption requires its exact ITERATION-BUNDLE or immutable reference. Terminal selection shall create a separate immutable record over completed iterations.

22 Normative requirements 19–21

SCI-FRUIT-REQ-019: Completion, convergence and stopping

Iteration completeness, resource exhaustion, convergence, policy stop, failure, terminal selection and consumer fitness shall be distinct claims. A convergence proposition shall bind quantity/domain, metric/norm, scale/denominator, tolerance, direction, persistence/consecutive rule, missing/non-finite behavior, generation and state. Resource limits shall bind exact accounting/counters and consequences. Completion or a reached limit shall not imply convergence; convergence shall not imply correctness, unit response, unbiasedness or adequacy. The conditional core shall select no convergence test, tolerance, limit, stopping rule or fitness policy.

SCI-FRUIT-REQ-020: Comparable terminal selection and open-world use

Selection shall use one exact predeclared rule over completed candidates with data access, ties/failure and all nine terminal fields. Candidates shall share or undergo exact selector-owned conversion into common quantity/role, units/scale, frame/grid/domain, reference/gauge, response, support, method/generation and uncertainty/metric meaning. All cross-route/method/grouping/gauge/generation/support conversions belong to selector identity. Retain every candidate, inclusion/exclusion, conversion, tie, unavailable comparison/cause and terminal response/uncertainty. Field 9 shall hold only requested/linked downstream decision identities and exact statuses, or an explicit no-request statement; no closed-world future-use list is required. Equal shape/scalar or last/latest/lowest-residual/visual preference shall not imply selection.

SCI-FRUIT-REQ-021: Universal bundle and role-local hard dependencies

Base iteration/terminal completion shall require all universal and exact method-required roles with values/reconstruction or core-and-method-permitted statuses. No method shall waive core roles or hard dependencies for result, continuation, response/uncertainty or selection. Unknown remains unavailable, never zero/identity/bypass/omission. An unavailable optional companion permits only a narrower product allowed by both core and method. The exact FRUIT-owned decision establishes completion and immutable publication. Downstream use shall not participate in base completion; only an exact explicitly requested FRUIT product role may require a specified child, with that dependency restricted to that requested role. A bare result payload shall not replace its bundle.

23 Normative requirements 22–24

SCI-FRUIT-REQ-022: Producer ownership and SCI-VAL-congruent child records

FRUIT shall preserve exact producer quantity/calibration/response/support/admission and source authority while owning model/update/recurrence, stopping, local completion, selector and publication. Downstream use shall be an immutable child reference, never a parent mutation or new FRUIT iteration. For SCI-VAL-governed use, retain exact request, applicability, eligibility and realization states plus profile/source binding. Not-requested creates no eligibility proposition. Missing profile/unresolved scope may yield a realized accounting record with applicability_unknown and decision_unavailable; generic unavailable shall not be a SCI-VAL realization value. Incomplete/failed/not-produced artifacts are separate realization conditions. SCI-VAL may evaluate an exact approved use policy but shall not execute FRUIT, advance recurrence or select model/terminal.

SCI-FRUIT-REQ-023: Complete method-record gate

Every method-specific instance shall bind all twenty-one required classes: authority/lineage; route/grouping; input rule; target/model; construction/selection/acceptance/application; accumulation meaning; typed composition; additive/unavailable modes; operation graph; state by operation; support/missing data; update/transition; response; uncertainty; continuation; stochastic state; convergence; resources; terminal selection; persistence/reconstruction; completion/failure/publication. Each shall satisfy the complete scientific obligations of this core, including corrected fixed-map/defining-state-rerun responses, closed uncertainty-role grammar and law-relative NOI member meaning, terminal conversions, open-world linked use, request/producer branches and universal dependencies. Actual values, owner approval and required upstream authority are mandatory for numerical admission; filled fields alone do not pass the gate.

SCI-FRUIT-REQ-024: Predictions, missing premises and claim ceilings

This complete sole current r0.4 review-candidate normative core supersedes r0.3 as the review candidate; it is not owner approved, frozen or activated. No unstated historical sentence supplies a current obligation. The method and claim shall preserve all six stable assumptions and twenty conditional predictions, with their exact premises, consequences and missing-premise behavior. Missing authority or premises shall block only the affected claim, with typed status, cause, scope and provenance; they shall not be repaired by an invented algorithm, default or evidence. Document verification shall not claim numerical conformance, fidelity, coverage, recovered modes, convergence adequacy, downstream fitness, observational validation, performance, readiness, production suitability/authorization or Unity activity. Numerical development, qualification and policy recommendations retain separate owner decisions and independent-pointing evidence obligations.

24 Conditional predictions 1–7

Conditional deductions only. Arithmetic requires ASM-001; complete linearity requires ASM-003, and covariance ASM-004. Missing premises block the affected claim.

SCI-FRUIT-PRED-001: Zero applied feedback

If $m_k^{\text{applied}} = 0$ and $\Pi_k(0) = B_k(0) = 0$ with identical compatible state/support/reference/gauge/null-space facts, substitution gives $z_k = F_k(y_k^{\text{in}})$. A zero candidate/accepted model or coordinate zero does not establish this premise; non-zero-preserving offsets defeat the reduction. No unavailable measured mode becomes observed zero.

SCI-FRUIT-PRED-002: Identity residual procedure

If F_k is identity on compatible $Y_k = Z_k$ and $\Pi_k = B_k$ on the applied operand with identical support/reference/gauge arithmetic, removal and rejoin cancel to $z_k = y_k^{\text{in}}$. Missing compatibility blocks cancellation. Neither equality of model stages nor recovery of an unavailable measured mode follows.

SCI-FRUIT-PRED-003: Zero net linear model path

Under complete fixed linearity and $B_k = A_k \circ \Pi_k$ on the applied-model domain, $D_k = 0$ and $z_k = A_k y_k^{\text{in}}$ is independent of that operand within the fixed conditions. This follows directly from the operator difference; changing state or rules changes the premise.

SCI-FRUIT-PRED-004: Perfect applied operand

If $y_k^{\text{in}} = \Pi_k(m_{\text{ref}}^{\text{applied}})$, $m_k^{\text{applied}} = m_{\text{ref}}^{\text{applied}}$ and $F_k(0) = 0$ in the declared reference/gauge, then $\rho_k = 0$ and $z_k = B_k(m_{\text{ref}}^{\text{applied}})$. A perfect accepted model alone is insufficient. Rejoin proves no true sky or unit response; unavailable-mode content remains model/prior supported.

SCI-FRUIT-PRED-005: Transfer, output shift and bias

With identical input/operators under complete fixed linearity, subtracting results at $m_{\text{ref}}^{\text{applied}} + \delta m_k^{\text{applied}}$ and $m_{\text{ref}}^{\text{applied}}$ yields $D_k \delta m_k^{\text{applied}}$. D_k is the transfer operator; the resulting difference is the distinct output-shift product. Model mismatch is separate, not covariance or bias. Bias is the law-qualified expected shift only with exact estimand/reference/error-law/conditioning authority; otherwise it is unavailable while transfer/shift may exist. Accepted error first undergoes the exact accepted-to-applied transformation/dependence.

SCI-FRUIT-PRED-006: Fixed-subspace projection and frozen-component subtraction

For exact frozen PTC operator-defining state on the admitted finite/rank/support domain, substitution gives $z_{g,t} = x_{g,t} P_{g,t,\Theta}^{\text{PTC}}$ and $\delta z_{g,t} = \delta x_{g,t} P_{g,t,\Theta}^{\text{PTC}}$. The internal modal coordinate changes algebraically with x under the same map, without RF-03 or operator-generation change. Fixed centering gives the affine map $(Y_{g,t}^{\text{CAL}} - \lambda_g)P$ with derivative HP , retaining its offset. Frozen-component subtraction instead has identity input derivative. Relearning/re-resolving a defining fact is a separately scoped procedure query; incidental equal outputs do not equate these families or authorize a route.

SCI-FRUIT-PRED-007: Model/prior origin is not exposure

Where a method permits model-only output or model contribution in an unavailable measured additive mode, that content retains model-or-prior-supported origin. It is not measured, recovered from data, acquired exposure or unit-response truth. Its ancestry is model/prior; neither rejoin nor a gauge choice changes that authority. Model-only permission itself must be supplied.

25 Conditional predictions 8–13

SCI-FRUIT-PRED-008: Support and additive incompatibility

Unequal support or incompatible reference/gauge/null-space facts permit a narrower occurrence class only with exact method permission and compatible arithmetic; otherwise affected composition is unavailable. A partial or incompatible map has no licensed sum there. Implicit filling, renormalization, offsets or gauge alignment do not repair this premise.

SCI-FRUIT-PRED-009: Fixed-map derivative and defining-state reruns

On compatible differentiable fixed-state tangent spaces the chain rule gives $J_{z \leftarrow y}^{\text{fixed}} = DF|_{\rho}$ and $J_{z \leftarrow m}^{\text{fixed}} = DB|_m - DF|_{\rho} \circ D\Pi|_m$. Internal fixed-map coordinates may change while this derivative applies. Complete fixed linearity reduces it to A_k, D_k ; affine offsets otherwise remain. Prescribed reruns that can change defining maps/state belong to RF-03; fixed-extent propagation to RF-04, stopping/selection to RF-05 and complete parent-domain composition to RF-07. A fixed derivative omits those causal reruns and cannot replace their query.

SCI-FRUIT-PRED-010: Fixed-domain failure and response boundary crossing

A perturbation leaving the exact finite/rank/support domain, or crossing a branch/reference/selection boundary, may make a fixed response unavailable. It does not silently become RF-03. A finite-difference/state-transition object requires its own compatible comparison and convention. Re-resolving an operator-defining fact is a declared procedure response; evaluating an internal coordinate under an unchanged fixed map is not. Freezing a prescribed defining-state rerun or rerunning a frozen defining fact changes the query. Failed differentiability/comparison cannot be repaired by substituting a fixed-branch derivative.

SCI-FRUIT-PRED-011: Law-relative joint covariance

Complete fixed linearity and finite compatible joint moments yield all four covariance terms under the exact declared law. A physical/random-parent target retains random parent/model values. An explicitly fixed realized-model target has zero model/cross blocks only within its narrower law and omits learning variation. A conditional NOI-assignment law may instead condition on realized parent, fixed model/operators and exact successful design resolution; it is not the physical-parent target. Accepted uncertainty transforms through exact dependence, including transformation uncertainty. Deterministic full-product offsets change the center, not the corresponding zero-target-law noise covariance.

SCI-FRUIT-PRED-012: Repeated information remains dependent

Reusing measured parents or shared model/calibration/reference information preserves ancestry and dependence. Iterations do not become independent measurements and imply no $1/\sqrt{\text{iteration count}}$ uncertainty scaling. Such a scaling would require an independently authorized population and stochastic premise, which iteration count does not provide.

SCI-FRUIT-PRED-013: Exact continuation

After completed absolute zero-based iteration N , continuation-sufficient S_{N+1} and identical future controls/compatible method/profile imply continuation at absolute index $N + 1$ agrees with uninterrupted scientific execution. Both consume the same causal state, including reference/gauge and stochastic state. N is an index, not a count; omitting a consequential fact defeats the premise. This deduction performs no replay.

26 Conditional predictions 14–17

SCI-FRUIT-PRED-014: Request generations, restart and early outcomes

A request generation has immutable not-requested or requested state; later request creates a new generation. Restart creates a new lineage. New realized values under unchanged learning rules create state/application generations; changed rules require method identity. Retry preserves parent state, attempt/randomness and earlier failure. Availability and producer outcome attach to the stage actually reached. An early stop fabricates no later applied, realized, completion or publication record. An allowed unavailable optional companion may coexist with a narrower core-and-method-permitted complete product.

SCI-FRUIT-PRED-015: Resource exhaustion is not convergence

Reaching a declared iteration/resource maximum establishes its accounting and stopping state only. It supplies none of the separate convergence proposition, metric, persistence or adequacy premises. Completion, limit, convergence and consumer fitness therefore cannot replace each other.

SCI-FRUIT-PRED-016: Comparable immutable terminal selection

Selecting completed absolute iteration N requires an exact rule and all nine fields. Candidates share common quantity/units/frame/domain/gauge/response/support/method-generation/metric meaning or exact selector-owned conversions. Shape/scalar agreement is insufficient; missing hard comparisons block affected selection. Retain every candidate, conversion, inclusion/exclusion, tie, unavailable comparison/cause and terminal response/uncertainty. Field 9 contains only requested/linked use decisions with exact statuses or a no-request statement; no exhaustive future-use list is needed. Selection creates an immutable relation and never mutates the selected bundle.

SCI-FRUIT-PRED-017: Grouping does not commute by implication

Observation-local, coadd-level and jointly learned groupings do not imply $\text{FRUIT}(\text{Coadd}(y_o)) = \text{Coadd}(\text{FRUIT}(y_o))$. Their learning, support and dependence may differ. Equality requires exact compatible operators/state/support and learning premises; this is absence of implied equivalence, not universal inequality or a numerical coaddition rule. Here y_o denotes only the compared observation inputs.

27 Conditional predictions 18–20

SCI-FRUIT-PRED-018: NOI target-law zero and full-parent members

Under complete fixed linearity with fixed model/state, full members produce $A_k y_{k,b}^{\text{full}} + D_k m_k^{\text{applied}}$ and their exact law-qualified center. A member with $\mathbb{E}_{\mathcal{D}_{k,g}^{\text{NOI}}}[\delta y_{k,b}^{\text{NOI}} | \mathcal{H}_{k,g}^{\text{NOI}}] = 0$ produces $A_k \delta y_{k,b}^{\text{NOI}}$, without deterministic rejoin. This zero is a target-law expectation: source/residual structure and nonzero spatial/member sample means may remain; no finite-ensemble centering or $B - 1$ correction follows. Full-affine center/estimator and exact source law need separate authority. Fixed, dependent-successor and relearned methods/ensembles retain distinct bindings; no numerical route or pooling is implied.

SCI-FRUIT-PRED-019: No numerical admission from a family

Without a complete exact separately owner-approved method-specific FRUIT-FEEDBACK-METHOD record and required upstream numerical authority, no numerical route/result is admitted. Creating the family, authoring/freezing the conditional core or verifying documents supplies neither approval nor conformance. All numerical routes remain `unavailable_pending_separate_owner_approval`.

SCI-FRUIT-PRED-020: Bundle-bound completion and downstream accounting

A bare ITERATION-RESULT payload is not a scientific handoff without its exact ITERATION-BUNDLE or immutable reference. Base local completion survives absent/failed downstream use; only an explicitly requested product role may require a specified child. A not-requested SCI-VAL use creates no eligibility proposition. Missing profile/scope can produce a realized accounting record with `applicability_unknown` and `decision_unavailable`, distinct from incomplete/failed/not-produced artifact realization.

Requested/linked decisions preserve exact profile/source/status bindings and immutable FRUIT parents without a new iteration generation. No actual numerical completeness or fitness is established here.